Question 1

a) Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set that contains the following subsets:

$A = \{x |\ x \text{ is a prime number}\};$
$B = \{x |\ x \text{ is a positive integer and } x^2 < 25\};$
$C = \{x |\ x \text{ is a multiple of } 5\}.$

(i) List all elements of sets A, B and C. (3 marks)

(ii) Solve the following:

(1) $(A \cap \bar{B}) \oplus C$. (3 marks)

(2) $A - C$. (1 mark)

(3) $\overline{B \cup C}$. $= \bar{B} \cap \bar{C}$ (2 marks)

(i) $A = \{2, 3, 5, 7\}$

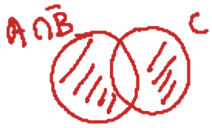
$B = \{1, 2, 3, 4\}$

$C = \{5, 10\}$

(ii) (1) $\bar{B} = \{5, 6, 7, 8, 9, 10\}$

$A \cap \bar{B} = \{5, 7\}$

$(A \cap \bar{B}) \oplus C = \{7, 10\}$



(2) $A - C = \{2, 3, 7\}$

(3) $B \cup C = \{1, 2, 3, 4, 5, 10\}$

$\overline{B \cup C} = \{6, 7, 8, 9\}$

b) Given $a = 2921$ and $b = 460$.

(i) Use the Euclidean algorithm to find the Greatest Common Divisor (GCD) of a and b. (3 marks)

(ii) Write the Greatest Common Divisor (GCD) of a and b in the form of $sa + tb$, where $s, t \in \mathbb{Z}$. (3 marks)

(iii) Hence, compute the Least Common Multiple (LCM) of a and b. (2 marks)

(i) $2921 = 460(6) + 161$

$460 = 161(2) + 138$

$161 = 138(1) + 23$

$138 = 23(6) + 0$

$\gcd(2921, 460) = 23$

$161 = 2921 - 460(6)$

$138 = 460 - 161(2)$

$23 = 161 - 138$

(ii) $23 = 161 - 138$

$= 161 - [460 - 161(2)]$

$= 161(3) - 460$

$= [2921 - 460(6)](3) - 460$

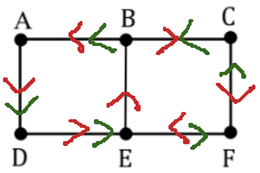
$= 2921(3) - 460(18) - 460$

$= \overset{a}{2921}(3) - \overset{b}{460}(19)$

$= (3)\,2921 + (-19)\,460$

(iii) $\operatorname{lcm}(2921, 460) = \dfrac{2921 \times 460}{23} = 58420$

c) Consider the graph below:



(i) Write the degree of each vertex of the graph. (2 marks)

(ii) Identify whether the graph contains an Euler circuit. If it does, write down a Euler circuit by listing the vertices in the order they are visited, starting and ending at vertex A. If it does not, explain why. (2 marks)

(iii) Identify whether the graph contains an Euler path. If it does, write down a Euler path by listing the vertices in the order they are visited, starting at vertex B. If it does not, explain why. (2 marks)

(iv) Identify whether the graph has a Hamiltonian circuit. If it does, write down a Hamiltonian circuit by listing the vertices in the order they are visited, starting and ending at vertex C. If it does not, explain why. (2 marks)

(i)

vertex	A	B	C	D	E	F
degree	2	3	2	2	3	2

(ii) No Euler circuit because vertices B and E have odd degree

(iii) Euler path exists.

Euler path: B, C, F, E, B, A, D, E

(iv) Hamiltonian circuit exists.

Hamiltonian circuit: C, B, A, D, E, F, C

Question 2

a) Let $A = \{a, b, c, d\}$ and the relation R on the set A be defined by

$$R = \{(a,a),(a,c),(a,d),(b,b),(b,d),(c,b),(c,d),(d,c),(d,d)\}.$$

(i) Identify the domain and range of the relation R. (2 marks)

(ii) Construct the matrix representation of R. (2 marks)

(iii) Compute the in-degree and out-degree of each vertex. (2 marks)

(i) $Dom(R) = \{a, b, c, d\}$

$Ran(R) = \{a, b, c, d\}$

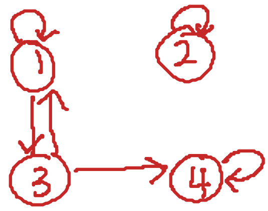
(ii) $$M_R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(iii)

vertex	a	b	c	d
in-degree	1	2	2	4
out-degree	3	2	2	2

b) Let $A = \{1, 2, 3, 4\}$ and the relation R on the set A be represented by the matrix

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$



(i) Examine whether the relation R is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, or transitive. Give a counterexample if the answer is "No".

(12 marks)

(ii) Identify whether R is an equivalence relation on A. Justify your answer.

(2 marks)

(iii) Apply Warshall's algorithm to obtain the matrix of the transitive closure of R.

(5 marks)

(i) R is not reflexive since $(3,3) \notin R$

R is not irreflexive since $(1,1) \in R$

R is not symmetric since $(3,4) \in R$ but $(4,3) \notin R$

R is not asymmetric since $(1,1) \in R$

R is not antisymmetric since $(1,3) \in R$ and $(3,1) \in R$ but $1 \neq 3$

R is not transitive since $(1,3) \in R$ and $(3,4) \in R$ but $(1,4) \notin R$

(ii) R is not an equivalence relation since R is not reflexive, not symmetric and not transitive.

(iii) Let $W_0 = M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

C_1: 1, 3

R_1: 1, 3

Add: $(1,1), (1,3), (3,1), (3,3)$

$W_1 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

C_2: 2

R_2: 2

Add: $(2,2)$

$W_2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

C_3: 1, 3

R_3: 1, 3, 4

Add: $(1,1), (1,3), (1,4), (3,1), (3,3), (3,4)$

$$W_3 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

C_4: 1, 3, 4

R_4: 4

Add: (1,4), (3,4), (4,4)

$$W_4 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = M_{R^\infty}$$

Question 3

a) Let $A = \{1, 2, 3, 4\}$ and let the relations R and S on A be represented by the following matrices:

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \text{ and } M_S = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}.$$

Compute:

(i) $M_{\overline{S}}$. (1 mark)

(ii) $M_{R^{-1} \cap S}$. (2 marks)

(iii) $M_{(R \cup S)^{-1}}$. (2 marks)

(iv) $M_{\overline{R \circ S}}$. (3 marks)

(i) $M_{\bar{S}} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$

(ii) $M_{R^{-1} \cap S} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \cap \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$

(iii) $M_{(R \cup S)^{-1}} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$

(iv) $M_{R \circ S} = M_S \odot M_R$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$M_{\overline{R \circ S}} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$1 \oplus 1 = 0$
$0 \oplus 1 = 1$
$1 \oplus 0 = 1$
$0 \oplus 0 = 0$

b) Consider the (2,5) encoding function $e: B^2 \to B^5$ defined by:

$$e(00) = 00000 = C_0$$
$$e(01) = 01100 = C_1$$
$$e(10) = 10101 = C_2$$
$$e(11) = 11001 = C_3$$

$C_2 \oplus C_3$

$$\begin{array}{r} 10101 \\ \oplus\ 11001 \\ \hline 01100 \end{array}$$

$C_1 \oplus C_1$

$$\begin{array}{r} 01100 \\ \oplus\ 01100 \\ \hline 00000 \end{array}$$

(i) Let $G = \{C_0, C_1, C_2, C_3\}$ where $C_0 = 00000$, $C_1 = 01100$, $C_2 = 10101$, and $C_3 = 11001$. Complete the table below.

$\oplus$	$C_0 = 00000$	$C_1 = 01100$	$C_2 = 10101$	$C_3 = 11001$
$C_0 = 00000$	$C_0 = 00000$	$C_1 = 01100$	$C_2 = 10101$	$C_3 = 11001$
$C_1 = 01100$	$C_1 = 01100$	$C_0 = 00000$	$C_3 = 11001$	$C_2 = 10101$
$C_2 = 10101$	$C_2 = 10101$	$C_3 = 11001$	$C_0 = 00000$	$C_1 = 01100$
$C_3 = 11001$	$C_3 = 11001$	$C_2 = 10101$	$C_1 = 01100$	$C_0 = 00000$

(4 marks)

$C_1 \oplus C_2$

$$\begin{array}{r} 01100 \\ \oplus\ 10101 \\ \hline 11001 \end{array}$$

$C_1 \oplus C_3$

$$\begin{array}{r} 01100 \\ \oplus\ 11001 \\ \hline 10101 \end{array}$$

(ii) Show that the (2,5) encoding function is a group code. (4 marks)

(iii) Compute the minimum distance of the group code and determine the number of error(s) that can be detected. (2 marks)

(ii) $C_i \oplus C_j \in G$ for $i = 0, 1, 2, 3$ and $j = 0, 1, 2, 3$

The identity is $C_0 = 00000 \in G$

$C_0^{-1} = C_0 \in G$

$C_1^{-1} = C_1 \in G$

$C_2^{-1} = C_2 \in G$

$C_3^{-1} = C_3 \in G$

$\therefore$ It is a group code.

(iii)

codeword	$C_1 = 01100$	$C_2 = 10101$	$C_3 = 11001$
weight	2	3	3

minimum distance = 2

can detect 1 error.

c) Given $A = \{1, 2, 3, 4, 5, 6\}$. Let $p_1 = (1, 3, 6, 4)$ and $p_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 3 & 4 & 5 & 2 \end{pmatrix}$ be permutations of A.

(i) Compute p_2^{-1}. (1 mark)

(ii) Compute $p_1 \circ p_2$ and write the result as a product of disjoint cycles and as a product of transpositions. (4 marks)

(iii) Determine whether permutation $p_1 \circ p_2$ is even or odd. Justify your answers. (2 marks)

(i) $P_2^{-1} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 3 & 4 & 5 & 1 \end{pmatrix}$

(ii) $P_1 \circ P_2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 6 & 1 & 5 & 4 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 1 & 3 & 4 & 5 & 2 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 3 & 6 & 1 & 5 & 2 \end{pmatrix}$

product of disjoint cycles $= (1, 4) \circ (2, 3, 6)$

product of transpositions $= (1, 4) \circ (2, 6) \circ (2, 3)$

(iii) It is odd permutation since number of transpositions is 3.

Question 4

a) Let $A = (\sim p \lor q) \leftrightarrow (r \rightarrow p)$.

(i) Complete the truth table given below.

p	q	r	$\sim p$	$\sim p \lor q$	$r \rightarrow p$	$(\sim p \lor q) \leftrightarrow (r \rightarrow p)$
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	F	T	F
T	F	F	F	F	T	F
F	T	T	T	T	F	F
F	T	F	T	T	T	T
F	F	T	T	T	F	F
F	F	F	T	T	T	T

(4 marks)

(ii) Identify whether the expression A is a tautology, contradiction or contingency.

(1 mark)

(iii) Find the Principal Disjunctive Normal Form (PDNF) and Principal Conjunctive Normal Form (PCNF) of A and $\sim A$.

(8 marks)

(ii) contingency

(iii) PDNF of $A = pqr + pq\bar{r} + \bar{p}q\bar{r} + \bar{p}\bar{q}\bar{r}$

PDNF of $\sim A = p\bar{q}r + p\bar{q}\bar{r} + \bar{p}qr + \bar{p}\bar{q}r$

PCNF of $A = \sim(\text{PDNF of } \sim A)$

$= (\bar{p}+q+\bar{r})(\bar{p}+q+r)(p+\bar{q}+\bar{r})(p+q+\bar{r})$

PCNF of $\sim A = \sim(\text{PDNF of } A)$

$= (\bar{p}+\bar{q}+\bar{r})(\bar{p}+\bar{q}+r)(p+\bar{q}+r)(p+q+r)$

b) Simplify $[(a \vee b)' \vee (a \wedge b')] \wedge b$ to its simplest form using the laws of Boolean algebra. (5 marks)

$$[(a \vee b)' \vee (a \wedge b')] \wedge b$$

$$= [(a' \wedge b') \vee (a \wedge b')] \wedge b$$

$$= [(a' \vee a) \wedge b'] \wedge b$$

$$= (1 \wedge b') \wedge b$$

$$= b' \wedge b$$

$$= 0$$

c) Let $f(x,y,z,w) = (x' \wedge y \wedge z \wedge w) \vee (x' \wedge y' \wedge z' \wedge w') \vee (x \wedge y \wedge z \wedge w) \vee (x \wedge y \wedge z \wedge w') \vee (x \wedge y' \wedge z' \wedge w') \vee (x' \wedge y \wedge z \wedge w')$.

(i) Construct a Karnaugh map using the form given below.

	z'	z'	z	z	
x'	1				y'
x'			1	1	y
x			1	1	y
x	1				y'
	w'	w	w	w'	

(5 marks)

(ii) Hence, simplify $f(x,y,z,w)$ to its simplest form using the Karnaugh map.

(2 marks)

$f(x,y,z,w) = yz + y'w'z'$